

Two Follow-up Questions: Gambler's Ruin under a Modified Winning Condition, and Bias / Node-Insertion in PageRank

Navansh Bansal

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1 Gambler's Ruin: Winning as "Any Profit"

1.1 Classical Setup & Bold Play

Consider an even-money game with win probability p and loss probability $q = 1 - p$. A gambler starts with fortune i and aims to reach a goal G before ruin at 0.

According to the **Dubins–Savage Theorem** [3], if the game is **sub-fair** ($p < q$), the optimal strategy to maximize survival is **Bold Play**. The bold strategy minimizes exposure to the negative expected drift by staking:

$$b(x) = \min\{x, G - x\}$$

This rule splits at the half-way mark $G/2$:

- $x < G/2$: Stake the entire fortune (Go all-in).
- $x \geq G/2$: Stake exactly the shortfall to reach the goal.

1.2 The Modified Goal: $G = i + 1$

If the objective shifts from reaching a distant target N to simply walking away with *any* non-zero profit, a profit of +2 is worth no more than +1. The effective goal collapses to $G = i + 1$.

Because the starting fortune $i \geq 1$ is always greater than or equal to $G/2$, the gambler is entirely in the "bet just enough to finish" branch. The initial stake is:

$$b(i) = (i + 1) - i = 1$$

While this looks like a Timid strategy bet, **Bold Play here is a doubling recovery**. If the first 1-unit bet loses, the fortune drops to $i - 1$. The new stake becomes $b(i - 1) = (i + 1) - (i - 1) = 2$.

Successive losses yield stakes of $1, 2, 4, 8, \dots$, pivoting to an all-in bet only if the fortune drops below the $(i + 1)/2$ threshold.

Takeaway. Redefining “winning” as any profit does **not** make timid play optimal. Bold play remains optimal in a sub-fair game. Because the start is just below the goal, bold play manifests as a small opening bet followed by escalating stakes to recover losses, capitalizing on the initial capital cushion.

1.3 Comparing Probabilities: Bold vs. Timid

Let $r = q/p > 1$. Assuming a starting fortune i , the success probabilities diverge significantly depending on the strategy:

(T) Timid.

Bet 1 continually until $i + 1$ or 0. The probability is exactly $\frac{r^i - 1}{r^{i+1} - 1}$, which asymptotically plateaus at p/q as $i \rightarrow \infty$.

(B) Bold.

Escalates $1, 2, 4, 8, \dots$ to recover. The massive cushion allows for roughly $\log_2 i$ consecutive losses before ruin. As $i \rightarrow \infty$, the success probability approaches 1.

Start i	$p = 0.40$		$p = 0.49$	
	Bold (B)	Timid (T)	Bold (B)	Timid (T)
4	0.679	0.616	0.789	0.784
10	0.821	0.663	0.902	0.890
100	0.965	0.667	0.989	0.960

1.4 Caveat: Winning Frequently $\neq$ Winning on Average

It is vital to distinguish between the *probability* of winning and the *mathematical expectation* of the game.

Bold play successfully maximizes how *often* you walk away with a profit. However, it cannot overcome the house edge. In a sub-fair game ($p < q$), your expected return is always negative.

The strategy simply trades magnitude for frequency. You achieve a high win rate by risking your massive starting capital to win just 1 unit. For example, if you start with $i = 10$ and $p = 0.40$, you will successfully reach 11 about 82% of the time. But in the remaining 18% of cases, you lose the entire 10.

Because $\mathbb{E}[\text{final}] = (11 \times 0.82) + (0 \times 0.18) \approx 9.02$, your expected final fortune is strictly less than your starting 10. The frequent, tiny wins are entirely wiped out by rare, catastrophic losses—a phenomenon often called “picking up pennies in front of a steamroller.”

2 PageRank: Sparse Data and Redirect Attenuation

2.1 Handling Transition Bias and Sparse Connectivity

A common intuitive model for a random surfer’s transition probabilities relies on empirical visit counts (MLE estimator): $\hat{p}_{ij} = n_{ij}/n_i$. While this correctly maps traffic [2], it suffers from extreme variance for rarely visited pages.

Classic Google PageRank [1] bypasses visit counts entirely, calculating base transition probabilities p_{ij} strictly from the static hyperlink out-degree (L_i) of page i . This is explicitly formulated as a two-case piecewise function:

$$p_{ij} = \begin{cases} \frac{1}{L_i} & \text{if page } i \text{ links to page } j, \\ 0 & \text{otherwise.} \end{cases}$$

However, this basic matrix faces a structural problem: dangling nodes (zero out-degree, where $L_i = 0$ leaves the transition undefined) and spider traps. Google resolves this using the **Google Matrix**:

$$G = \alpha S + (1 - \alpha) \frac{1}{N} \mathbf{1}\mathbf{1}^\top, \quad \alpha \approx 0.85 \quad (1)$$

Statistically, the teleportation vector $\frac{1-\alpha}{N}$ functions identically to **Laplace / Dirichlet additive smoothing** with a uniform prior. By bounding every transition probability strictly above zero, it regularizes isolated graph structures and guarantees a unique stationary distribution.

2.2 The Rank-Attenuating Effect of Redirect Nodes

Consider interposing a dedicated redirect page k between i and j ($i \rightarrow k \rightarrow j$), where k links only to j with a transition probability of 1.

Using the equilibrium PageRank formula, the new page k receives:

$$\text{PR}(k) = \frac{1 - \alpha}{N'} + \alpha \frac{\text{PR}(i)}{L_i} \quad (2)$$

Because $L_k = 1$, page k passes its rank to j :

$$\underbrace{\alpha \text{PR}(k)}_{\text{New contribution to } j} = \underbrace{\alpha^2 \frac{\text{PR}(i)}{L_i}}_{\text{Propagated part}} + \frac{\alpha(1 - \alpha)}{N'} \quad (3)$$

Originally, j received $\alpha \text{PR}(i)/L_i$ directly. By interposing k , the primary propagated rank is multiplied by an additional factor of $\alpha \approx 0.85$.

Takeaway. In the Markov framework, each intermediate node acts as a leaky pipe, bleeding roughly 15% of the propagated rank to the global teleportation vector. Therefore, a probability-1 redirect strictly *dilutes* the target's rank. (Note: Google's engineering engine artificially patches this by canonicalizing 30x redirects [4], passing link equity through without the Markov α -penalty).

References

- [1] L. Page, S. Brin, R. Motwani, and T. Winograd, *The PageRank citation ranking: Bringing order to the web*, Stanford InfoLab Technical Report, 1999.
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